

3. SEQUENCE OPERATIONS AND SIGNAL ANALYSIS

Objective

This experiment is divided into four simulations:

1. Addition and Multiplication of Two Sequences
2. Energy and Power Calculation of Given Signals
3. Autocorrelation
4. Cross Correlation

Experiment 3.1 :- Addition and Multiplication of Two Sequences

Program:

```
clc;
clear;
close all;
% Sample index
n = 0:10;
% Sequence 1
x1 = [1 2 3 4 5 6 7 8 9 10 11];
% Sequence 2
x2 = [11 10 9 8 7 6 5 4 3 2 1];
% Addition of sequences
y_add = x1 + x2;
% Multiplication of sequences
```

```
y_mul = x1 .* x2;
% Plotting
figure;
subplot(2,2,1);
stem(n,x1,'filled');
grid on;
title('Sequence x_1(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,2);
stem(n,x2,'filled');
grid on;
title('Sequence x_2(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,3);
stem(n,y_add,'filled');
grid on;
title('Addition: x_1(n) + x_2(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,4);
stem(n,y_mul,'filled');
grid on;
```

```
title('Multiplication: x_1(n) * x_2(n)');
```

```
xlabel('n');
```

```
ylabel('Amplitude');
```

Expected Output:

When you run the MATLAB code, you should observe four subplots:

1. Sequence 1 (Sinusoidal Sequence):

- o Characteristics: The first subplot shows a sinusoidal sequence with 30 samples. The sequence oscillates between positive and negative values, reflecting a sinusoidal waveform.

- o Description: The sequence is periodic with a frequency of 0.05 cycles/sample.

2. Sequence 2 (Exponentially Decaying Sequence):

- o Characteristics: The second subplot shows an exponentially decaying sequence. The sequence starts at a higher value and decays towards zero as the sample index increases.

- o Description: The sequence is aperiodic and decreases in amplitude with time.

3. Addition of Sequences:

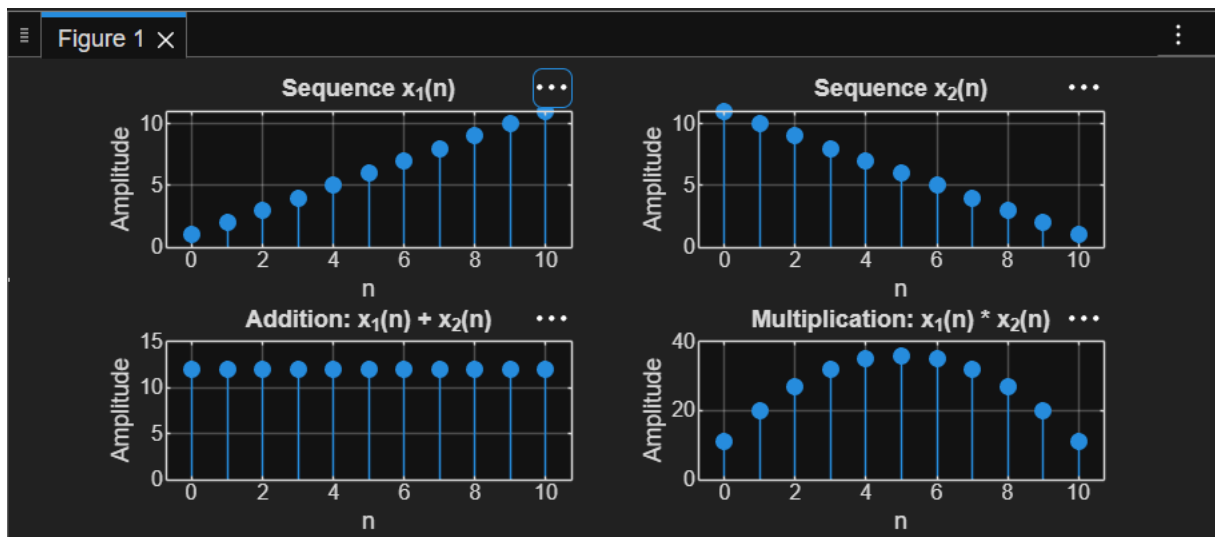
- o Characteristics: The third subplot shows the result of adding Sequence 1 and Sequence 2. The resultant sequence combines the oscillatory nature of the sinusoid with the decaying trend of the exponential sequence.

- o Description: The sequence's overall amplitude is influenced by both sequences, resulting in a waveform that decays over time but retains the sinusoidal shape.

4. Multiplication of Sequences:

- o Characteristics: The fourth subplot shows the result of multiplying Sequence 1 and Sequence 2. The resulting sequence has a waveform that is both oscillatory and exponentially decaying.
- o Description: The sequence's amplitude decreases over time due to the multiplication by the exponentially decaying sequence, while the sinusoidal pattern is maintained

Output Waveform:



Result:

Two discrete-time sequences were generated and their addition and point-wise multiplication

were performed successfully using MATLAB. The resulting sequences were displayed using

stem plots, illustrating basic operations on discrete-time signals.

Experiment 3.2 Energy and Power Calculation of Given Signals

Program

% Energy and Power Calculation of Signals

clc;

clear;

close all;

% Time index

n = -10:10;

% Signal

x = [zeros(1,10) 1 2 3 4 5 4 3 2 1 zeros(1,10)];

% Energy Calculation

E = sum(abs(x).^2);

% Power Calculation

P = E/length(x);

% Display Results

disp(['Energy of the signal = ', num2str(E)]);

disp(['Power of the signal = ', num2str(P)]);

% Plot Signal

figure;

stem(n,x,'filled');

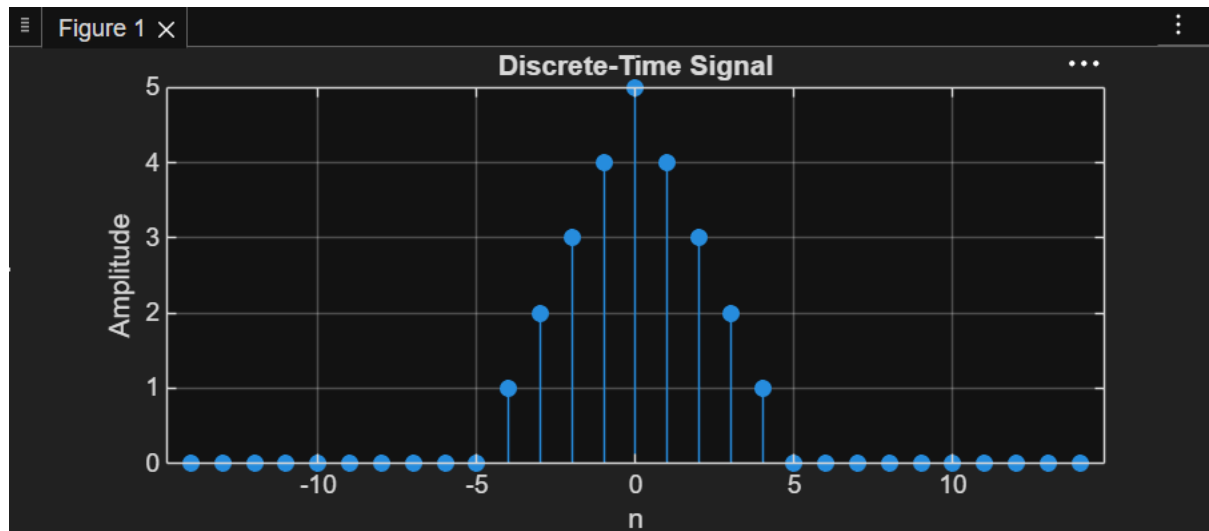
grid on;

title('Discrete-Time Signal');

xlabel('n');

```
ylabel('Amplitude');
```

Output Generated Graph :



Result

The energy and average power of the given discrete-time signal were calculated successfully using MATLAB. The signal energy was found to be **85**, and the average power was **2.9310**.